SKYRUN CUSTOMER ANALYSIS

Business Analysis and Recommendations Report

Prepared as part of the Amdari Data Science Project

1. Executive Summary

This report examines Skyrun customer purchasing behaviour, sales performance, customer retention and customer segmentation. The analysis aims to identify opportunities to improve customer engagement, retention and business revenue.

2. Dataset and Methodology

The analysis includes data cleaning, exploratory data analysis, cohort analysis, RFM analysis and customer segmentation. Customers were evaluated using purchasing recency, frequency and monetary value.

3. Cohort Analysis

Cohort analysis groups customers according to their first purchase month and measures their subsequent purchasing activity. Retention percentages help identify changes in customer engagement over time.

Cohort heatmap file was not found. Add the saved heatmap to the Charts folder before final submission.

4. Customer Segmentation

RFM analysis evaluates customers based on how recently they purchased, how frequently they purchase and how much they spend. Customer segmentation helps identify valuable customers and those who may require targeted engagement.

Segment	recency_days	frequency	monetary	Customer_Count
At Risk	153.39912917271408	3.985486211901306	147873.8797822932	689
Champions	4.745161290322581	22.7	1141111.6068064517	310
Lost Customers	216.7179245283019	1.2018867924528305	55691.9209245283	1060
Loyal Customers	14.86951219512195	8.163414634146342	351337.9399878049	820
Potential Customers	49.58924485125858	3.69908466819222	185456.59479405032	874
Recent Customers	16.17447495961228	1.948303715670436	117263.9887075929	619

5. Business Analysis Visualisations

6. Business Recommendations

1. Introduce targeted campaigns to encourage first-time customers to make repeat purchases.

2. Develop loyalty rewards and personalised offers for high-value customers.

3. Identify customers at risk of becoming inactive and implement win-back campaigns.

4. Use cohort retention results to improve customer engagement after the first purchase.

5. Use sales trends to support inventory and staffing planning.

7. Conclusion

The analysis provides a structured understanding of Skyrun's customer behaviour and purchasing patterns. Applying the recommended retention, engagement and customer segmentation strategies can help support better-informed business decisions.